DHIVYA BHARATHI S

AI/ML Trainee | Machine Learning | Deep Learning | Data Science | Analytics Consulting

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PROFESSIONAL SUMMARY

B.E. Biomedical Engineering graduate (CGPA 8.7/10, 2026, no active arrears) with a strong foundation in Python, SQL, Statistics, and Linear Algebra applied to Machine Learning and Deep Learning (CNN, Neural Networks, MobileNet). Hands-on experience developing, training, and deploying ML models using NumPy, Pandas, scikit-learn, and TensorFlow/Keras. Built and deployed end-to-end AI/ML systems including a Retrieval-Augmented Generation (RAG) application (LangChain, FAISS, Groq/Llama 3.1) and CNN-based image classification systems integrated with REST APIs, AWS Cloud, and IoT (ESP32). Three technical internships with hands-on Agile/QA and client-facing coordination experience. Seeking an AI/ML Trainee / Analyst role to apply data-driven problem solving to real-world business challenges.

TECHNICAL SKILLS

Programming & SQL: Python (OOP, Scripting, Automation), SQL (Joins, Aggregations, Subqueries), C, C++ (Basic), Data Structures & Algorithms

Machine Learning: NumPy, Pandas, scikit-learn, Statistics, Probability, Feature Engineering, Model Training & Evaluation, Data Analysis

Deep Learning: CNN, Neural Networks, MobileNet/MobileNetV2, TensorFlow/Keras, OpenCV, Computer Vision, LSTM (Basic)

LLM / GenAI: LangChain, FAISS, HuggingFace Sentence-Transformers, Groq API (Llama 3.1), Prompt Engineering, RAG Pipelines

Cloud & IoT: AWS (EC2, S3, IAM, CloudWatch, VPC), ThingSpeak, ESP32, ESP32-CAM, Sensor Integration

Tools & Practices: REST API (Flask), Git/GitHub, Streamlit, JSON, HTML5/CSS3/JavaScript, QA, SDLC, Agile/Scrum

KEY PROJECTS

Smart Veterinary Glove (VETFUSION) — AI Disease & Wound Infection Detection 2025–2026

Python | CNN | MobileNet | TensorFlow/Keras | OpenCV | AWS | REST API | ESP32-CAM | ThingSpeak

- Captured 3,001 ESP32-CAM wound images across 4 classes and trained a MobileNet-based CNN via transfer learning (80.22% validation / 76% test accuracy, F1: 0.74); integrated predictions with live sensor data in an Android app via ThingSpeak.

GitHub: github.com/dhivyabharathi781/Smart-glove-real-time-wound-infection-detection

AI Resume ATS Analyzer — RAG-based Resume-to-JD Matching Tool 2026

Python | LangChain | FAISS | HuggingFace Sentence-Transformers | Groq API (Llama 3.1) | Streamlit

- Built a RAG application comparing a resume against a JD to generate ATS-style feedback (match score, missing skills, suggestions); chunked and locally embedded resume text, indexed in FAISS for top-relevant retrieval.
- Prompted a Groq-hosted LLM (Llama 3.1) to return strict schema-validated JSON with defensive parsing, wrapped in a real-time Streamlit UI.

GitHub: github.com/dhivyabharathi781/AI-Resume-ATS-Analyzer

Cloud-Based Plant Disease Detection System 2026

Python | CNN | MobileNetV2 | TensorFlow/Keras | Flask | AWS (EC2, S3, IAM, CloudWatch)

- Trained custom CNN and MobileNetV2 models on the PlantVillage dataset; built a Flask REST API on AWS EC2 for real-time predictions, with S3 storage and CloudWatch monitoring.

GitHub: github.com/dhivyabharathi781/Plant_disease_detection

INTERNSHIP & WORK EXPERIENCE

IoT Intern → AI/ML Intern → Project Coordinator — LGTG Solutions Pvt. Ltd., Chennai (Part-time) Jan – Aug 2025

- Integrated ESP32 and sensor modules (temperature, humidity, color) for real-time IoT data collection; transmitted live sensor readings to ThingSpeak Cloud via REST API.
- Built Python (OOP) data pipelines using NumPy and Pandas for sensor data acquisition, preprocessing, and feature extraction to support ML model development.
- Coordinated a 5-member Agile sprint team and client-facing requirement discussions, driving QA checks, sprint planning, and technical documentation.

AI/ML Intern — Codec Technologies, Chennai (Remote) Dec 2024 – Jan 2025

- Implemented ML algorithms using Python and scikit-learn on sample datasets, applying statistical feature engineering techniques.

Cloud Computing Intern — YHills (Remote) Feb – Mar 2025

- Provisioned AWS resources (EC2, S3, IAM, CloudWatch) and configured IAM role-based access control; tested cloud data workflows for storage, retrieval, and monitoring.

EDUCATION

B.E. Biomedical Engineering — Dr. M.G.R. Educational and Research Institute, Chennai 2022 – 2026

CGPA: 8.7/10 | No Active Arrears/Backlogs

Relevant Coursework: Statistics & Probability, Signal Processing, Biomedical Instrumentation, Microprocessors & Microcontrollers, Analog & Digital Electronics

CERTIFICATIONS & ACHIEVEMENTS

Nestlé E-Learning 2026 – Resilience (Certificate of Completion, Nestlé "needs YOUth" Nesternship Program)

HackerRank Python (Basic) • HackerRank SQL (Basic) • Codec Technologies – AI & ML • NPTEL – IoT • NPTEL – Biomedical Nanotechnology • Simplilearn – Web Dev • YHills – Cloud Computing • 1st Prize – Technical Symposium • Best Promoter Award – UN SDG Initiative

LANGUAGES

English (Professional), Tamil (Native), Spanish (A1)